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Tutorial Course for Intro to Sensor-Integrated Systems

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Mechanical Deformation and Electromechanical Conversion

1. Core Physical Effects (Theoretical Basis)

- **Hooke's Law:** Force $\rightarrow$ causes mechanical deformation
- **Poisson Effect:** Longitudinal stretching is inevitably accompanied by transverse contraction (The core for transverse strain gauge calculations)
- **Strain Effect:** Metal wire stretches and becomes thinner $\rightarrow$ **Resistance increases (+)**; compresses and becomes thicker $\rightarrow$ **Resistance decreases (-)**

2. Core Formulas and Parameter Derivation

- **Normal Stress (Stress):**

$$\sigma = \frac{F}{A}$$

(Note: F is the axial force; tension is positive, compression is negative)

- **Longitudinal Strain:**

$$\varepsilon_L = \frac{\sigma}{E}$$

(Note: E is Young's modulus, representing material stiffness)

- **Transverse Strain:**

$$\varepsilon_T = -\nu \cdot \varepsilon_L$$

(Reminder of common mistake: ν is Poisson's ratio, the **negative sign** in the formula must absolutely not be omitted!)

- **Electromechanical Conversion (Relative change rate of resistance):**

$$\frac{\Delta R}{R} = G \cdot \varepsilon$$

(Note: G is the strain gauge sensitivity factor (gauge factor). Substitute ε_L for longitudinal mounting, substitute ε_T for transverse mounting)

Measurement Circuits — DC Wheatstone Bridge

1. Core Circuit Principles

- **Purpose:** The resistance change of a single strain gauge is extremely small, so a bridge circuit is required to convert it into a measurable voltage difference U_o .
- **Differential Principle:** Adjacent bridge arms "one under tension, one under compression" → output voltages are superimposed and amplified.
 - Adjacent bridge arms change in the same direction under the influence of ambient temperature → output voltages cancel each other out (achieving temperature compensation).

2. Core Formulas and Engineering Characteristics *(Assuming the initial resistance of all four arms is R , and the supply voltage is U_i)*

① Quarter Bridge - Only 1 bridge arm is a strain gauge

- **Formula:** $U_o \approx \frac{1}{4} \cdot \frac{\Delta R}{R} \cdot U_i$
- **Characteristics:** Sensitivity baseline, presence of non-linear error, **no** temperature compensation.

② Half Bridge (Differential) - 2 bridge arms are strain gauges (usually one tension and one compression connected to adjacent arms)

- **Formula:** $U_o \approx \frac{1}{2} \cdot \frac{\Delta R}{R} \cdot U_i$
- **Characteristics:** Sensitivity is **2 times** that of a quarter bridge, eliminates non-linear error, possesses self-temperature compensation.

③ Full Bridge (Differential) - All 4 bridge arms are strain gauges

- **Formula:** $U_o \approx \frac{\Delta R}{R} \cdot U_i$
- **Characteristics:** Sensitivity is **4 times** that of a quarter bridge, strongest anti-interference capability, the first choice for industrial applications!

Exercise

A DC strain gauge bridge circuit is shown in the figure. The bridge power supply is $E = 4V$.

The initial resistances of the four arms are equal: $R_1 = R_2 = R_3 = R_4 = 120\Omega$. When

subjected to strain, the absolute change in resistance for each active gauge is $\Delta R_1 =$

$\Delta R_2 = \Delta R_3 = \Delta R_4 = 1.2\Omega$.

Calculate the approximate output voltage ($U_o \approx ?$) for the following four conditions:

1. **Quarter Bridge:** Only R_1 is an active strain gauge, and the others are fixed resistors.
2. **Half Bridge (Differential):** R_1 and R_2 are active strain gauges experiencing **opposite** strains (one tension, one compression), and they are connected in adjacent arms.
3. **Half Bridge (Same Strain):** R_1 and R_2 are active strain gauges experiencing the **same** strain (both tension or both compression), and they are connected in adjacent arms.
4. **Full Bridge:** R_1, R_2, R_3 , and R_4 are all active strain gauges forming a differential full bridge circuit.

Solution

General Approximation Formula: When $\Delta R \ll R$, the output voltage U_o can be approximated as:

$$U_o \approx \frac{\Delta R_1 - \Delta R_2 + \Delta R_3 - \Delta R_4}{4R} E$$

1. Quarter Bridge

- Only $\Delta R_1 = 1.2\Omega$ exists; others are 0.

- $$U_o \approx \frac{\Delta R}{4R} E = \frac{1.2}{4 \times 120} \times 4 = 0.01V = \mathbf{10mV}$$

2. Half Bridge - Opposite Strains

- R_1 and R_2 are in adjacent arms. Tension gives $+\Delta R$, compression gives $-\Delta R$.

- $$U_o \approx \frac{\Delta R - (-\Delta R)}{4R} E = \frac{2\Delta R}{4R} E = \frac{2 \times 1.2}{4 \times 120} \times 4 = 0.02V = \mathbf{20mV}$$

3. Half Bridge - Same Strains

- Both experience tension ($+\Delta R$) or both experience compression ($-\Delta R$).

- $$U_o \approx \frac{\Delta R - \Delta R}{4R} E = \mathbf{0V}$$

4. Full Bridge

- Four gauges, adjacent arms have opposite strains, opposite arms have same strains.

- $$U_o \approx \frac{\Delta R - (-\Delta R) + \Delta R - (-\Delta R)}{4R} E = \frac{4\Delta R}{4R} E = \frac{4 \times 1.2}{4 \times 120} \times 4 = 0.04V = \mathbf{40mV}$$

Exercise

Description:

A load cell consists of a solid steel cylinder 5 cm in diameter and 5 cm long. An axial compressive load of 10^5 N is applied to the cylinder.

Given the material properties of steel:

- Young's modulus (E) = 2×10^{11} N/m² (or Pa)
- Poisson's ratio (ν) = 0.3

Find:

1. The longitudinal strain (ϵ_L)
2. The transverse strain (ϵ_T)

Solution

Step 1: Calculate the Cross-sectional Area (A)

The diameter of the cylinder is $d = 5 \text{ cm} = 0.05 \text{ m}$.

$$A = \frac{\pi \cdot d^2}{4} = \frac{\pi \cdot (0.05)^2}{4}$$

$$A \approx 1.963 \times 10^{-3} \text{ m}^2$$

Step 2: Calculate the Axial Stress (σ)

Since the applied force $F = 10^5 \text{ N}$ is a compressive load, we denote it as negative (-10^5 N).

$$\sigma = \frac{F}{A} = \frac{-10^5}{1.963 \times 10^{-3}}$$

$$\sigma \approx -5.093 \times 10^7 \text{ N/m}^2 \text{ (Pa)}$$

Step 3: Calculate the Longitudinal Strain (ϵ_L)

Using Hooke's Law:

$$\epsilon_L = \frac{\sigma}{E} = \frac{-5.093 \times 10^7}{2 \times 10^{11}}$$

$$\epsilon_L \approx -2.546 \times 10^{-4} \text{ (or } -254.6 \mu\epsilon)$$

Step 4: Calculate the Transverse Strain (ϵ_T)

Using Poisson's ratio:

$$\epsilon_T = -\nu \cdot \epsilon_L = -0.3 \times (-2.546 \times 10^{-4})$$

$$\epsilon_T \approx +7.639 \times 10^{-5} \text{ (or } +76.4 \mu\epsilon)$$

Exercise

Description:

Two strain gauges are bonded to the solid steel cylinder from the previous problem (Problem 8.17).

- **Gauge 1** is parallel to the cylinder axis (longitudinal direction).
- **Gauge 2** is at right angles to it (transverse direction).

The unstrained resistance of each gauge is $R = 120\Omega$ and the gauge factor is $G = 2.0$.

Recall from Problem 8.17:

- Longitudinal strain: $\epsilon_L = -2.546 \times 10^{-4}$
- Transverse strain: $\epsilon_T = +7.639 \times 10^{-5}$

Find:

Calculate the change in resistance (ΔR) of each gauge.

Solution

General Formula:

The relationship between the change in resistance and the mechanical strain is given by:

$$\frac{\Delta R}{R} = G \cdot \epsilon \quad \Rightarrow \quad \Delta R = R \cdot G \cdot \epsilon$$

Step 1: Calculate the resistance change for Gauge 1 (Longitudinal)

Gauge 1 is attached parallel to the axis, so it experiences the longitudinal strain (ϵ_L).

$$\Delta R_1 = R \cdot G \cdot \epsilon_L$$

$$\Delta R_1 = 120 \times 2.0 \times (-2.546 \times 10^{-4})$$

$$\Delta R_1 \approx -0.0611 \Omega$$

Step 2: Calculate the resistance change for Gauge 2 (Transverse)

Gauge 2 is attached at a right angle to the axis, so it experiences the transverse strain (ϵ_T)

caused by the Poisson effect.

$$\Delta R_2 = R \cdot G \cdot \epsilon_T$$

$$\Delta R_2 = 120 \times 2.0 \times (+7.639 \times 10^{-5})$$

$$\Delta R_2 \approx +0.0183 \Omega$$

Mechanical Analysis and Gauge Placement Strategy

1. Mechanical Characteristics of a Cantilever Beam

- **Force and Deformation:** When a downward force F is applied to the free end of the cantilever beam, bending stress is generated inside.
 - **Upper surface:** Bulges outward $\rightarrow$ under tension (Tension, $+\epsilon$) $\rightarrow$ strain gauge resistance increases ($+\Delta R$)
 - **Lower surface:** Curves inward $\rightarrow$ under compression (Compression, $-\epsilon$) $\rightarrow$ strain gauge resistance decreases ($-\Delta R$)
- **Stress Distribution Law:** The closer to the fixed end (root of the beam), the greater the bending strain. Therefore, strain gauges must be bonded close to the fixed end to obtain the maximum signal.

2. Bridge Wiring Golden Rules

- **Design Objective:** To make the resistance changes of all active strain gauges superimpose in the same direction in the bridge output.
- **General Approximate Formula of the Bridge:** $U_o \propto (\Delta R_1 - \Delta R_2 + \Delta R_3 - \Delta R_4)$
- **Core Mnemonics:**
 - **"One tension, one compression, connect to adjacent bridge arms"** (e.g., if R_1 is connected to the upper surface tension gauge, R_2 must be connected to the lower surface compression gauge; minus and minus make plus, doubling the signal)
 - **"Same tension / same compression, connect to opposite bridge arms"** (e.g., R_1 and R_3 are opposite to each other, both connected to the upper surface tension gauges)

Non-linear Error and Elimination

1. Non-linear Error of a Quarter Bridge

➤ Exact Formula of a Quarter Bridge:

$$U_o = \frac{\Delta R}{4R + 2\Delta R} U_i$$

➤ **Cause of Error:** The denominator contains the independent variable $2\Delta R$. When the external force increases and the strain becomes larger, the relationship between U_o and ΔR is no longer perfectly linear, causing the output curve to bend.

➤ Maximum Non-linear Error Percentage:

$$\gamma \approx \frac{2\Delta R}{4R} \times 100\% = \frac{\Delta R}{2R} \times 100\%$$

2. Differential Elimination

➤ Half Bridge Differential (One tension and one compression connected to adjacent arms):

$$U_o = \left(\frac{R + \Delta R}{(R + \Delta R) + (R - \Delta R)} - \frac{1}{2} \right) U_i = \frac{\Delta R}{2R} U_i$$

➤ Full Bridge Differential (Two tension and two compression cross-connected):

$$U_o = \frac{\Delta R}{R} U_i$$

➤ Core Conclusion and Engineering Significance:

- After adopting the differential half bridge or full bridge, the variable ΔR in the denominator is **completely canceled out**, and the denominator becomes a constant.
- **The non-linear error drops directly to 0%**, achieving a perfectly linear output, while the measurement sensitivity is simultaneously increased by 2 to 4 times!

Exercise

Description:

It is proposed to design a force measurement system using a cantilever beam of uniform cross-section. **Four identical resistance strain gauges** will be bonded to the beam to form a **differential full-bridge circuit**.

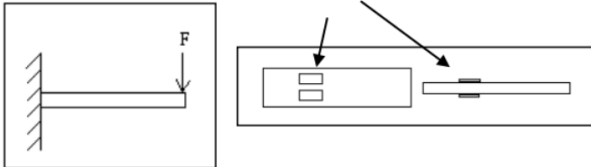
Questions:

1. **Gauge Placement:** How should these four strain gauges be attached to the cantilever beam? (Describe the placement and draw the schematic).
2. **Circuit Wiring:** How should they be connected in a Wheatstone bridge circuit to achieve maximum sensitivity and temperature compensation? (Draw the circuit diagram).

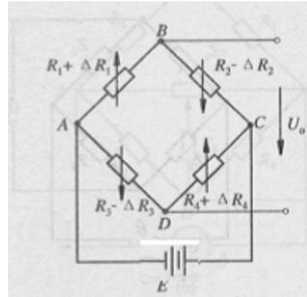
Solution

Part 1: Gauge Placement Strategy

- **Mechanical Analysis:** When a downward force F is applied to the free end of the cantilever beam, it bends.
 - The **upper surface** undergoes tensile strain (Tension, $+\epsilon$).
 - The **lower surface** undergoes compressive strain (Compression, $-\epsilon$).
 - The bending moment is maximum near the fixed end (root) of the beam.
- **Placement:** To obtain the maximum signal, all four gauges should be placed near the **fixed end**.
 - Two gauges (R_1 and R_3) must be bonded to the **top surface** to measure tension.
 - Two gauges (R_2 and R_4) must be bonded to the **bottom surface** to measure compression.



Part 2: Bridge Circuit Wiring



Exercise

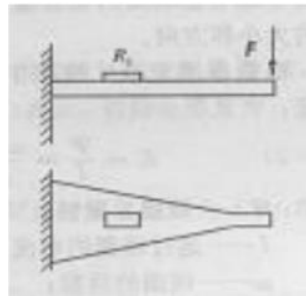
Description:

A force measurement system uses a cantilever beam. A single strain gauge R_1 is bonded to the surface of the beam. The unstrained resistance of the gauge is $R = 120\Omega$, and its gauge factor is $G = 2.0$.

The gauge is connected to a **quarter-bridge circuit (single active arm)**. The other three arms are fixed resistors of 120Ω . The bridge excitation voltage is $U_i = 5V$. When a load F is applied, the strain gauge experiences a longitudinal strain of $\epsilon = 1000\mu\epsilon$ (1000×10^{-6}).

Questions:

1. Calculate the exact output voltage (U_o) of this quarter-bridge circuit.
2. Calculate the non-linear error percentage of the measurement.
3. Explain how to modify the circuit design to reduce or eliminate this non-linear error.



Solution

Part 1: Calculate the Exact Output Voltage

- Relative resistance change:

$$\frac{\Delta R}{R} = G \cdot \epsilon = 2.0 \times (1000 \times 10^{-6}) = 0.002$$

- Absolute resistance change:

$$\Delta R = 0.002 \times 120\Omega = 0.24\Omega$$

- Exact output equation for a quarter bridge:

$$U_o = U_i \left(\frac{R + \Delta R}{(R + \Delta R) + R} - \frac{1}{2} \right) = U_i \left(\frac{\Delta R}{4R + 2\Delta R} \right)$$

- Calculation:

$$U_o = 5V \times \left(\frac{0.24}{4(120) + 2(0.24)} \right) = 5 \times \left(\frac{0.24}{480.48} \right) \approx \mathbf{0.0024975V} \text{ (2.4975mV)}$$

Part 2: Calculate the Non-linear Error

- Ideal Linear Output (Approximation):

$$U_{o(linear)} = U_i \left(\frac{\Delta R}{4R} \right) = 5V \times \left(\frac{0.24}{480} \right) = \mathbf{0.0025V} \text{ (2.5mV)}$$

- Non-linear Error Percentage (γ):

$$\gamma = \frac{|U_{o(linear)} - U_{o(exact)}|}{U_{o(linear)}} \times 100\%$$

Alternatively, using the theoretical error formula:

$$\gamma \approx \frac{\Delta R}{2R} \times 100\% = \frac{0.24}{240} \times 100\% = \mathbf{0.1\%}$$

Solution

Part 3: How to Eliminate the Error (System Optimization)

- **Solution:** Upgrade the circuit from a quarter-bridge to a **differential half-bridge** or **full-bridge**.
- **Explanation:** By bonding a second identical strain gauge on the opposite surface of the beam (experiencing compressive strain, $-\epsilon$) and connecting it to the **adjacent arm** of the bridge, the variable ΔR in the denominator is canceled out. This not only doubles the sensitivity but also mathematically forces the non-linear error to **0%**.

Column-type Integrated Force Measurement System

1. Mechanical Strategy (Core Challenge & Breakthrough)

- **Force Dilemma:** Heavy-duty load columns only withstand axial compression. Surface strain gauges can only detect compression signals, making it impossible to form a differential full bridge directly.
- **Engineering Breakthrough: Cleverly utilizing the Poisson Effect**
 - **Longitudinal Mounting:** Parallel to the axis of the applied force, sensing axial compression $\rightarrow$ resistance decreases $(-\Delta R)$
 - **Transverse Mounting:** Perpendicular to the axis of the applied force, sensing transverse expansion (getting thicker) $\rightarrow$ obtains a natural tension signal $(+\Delta R)$
 - **Conversion Core:** Transverse strain $\epsilon_T = -\nu \cdot \epsilon_L$ (ν is the Poisson's ratio of the material, typically around 0.3 for steel)

2. Cross-Dimensional Differential Full Bridge Integration (Bridge Integration)

- **Wiring Principle:** Adhering to the golden rule of "**one tension, one compression, connect to adjacent bridge arms**", the longitudinal gauges (compression) and transverse gauges (tension) are connected alternately into the bridge circuit.
- **Output Equation:**

$$U_o \approx \frac{\Delta R_{trans} - (-\Delta R_{long}) + \Delta R_{trans} - (-\Delta R_{long})}{4R} U_i = \frac{2(1 + \nu)\Delta R_{long}}{4R} U_i$$

- **Engineering Value:** Although the output is not exactly 4 times that of a quarter bridge due to Poisson's ratio (it is approximately 2.6 times), it **completely eliminates non-linear error** and achieves perfect self-temperature compensation.

3. Signal Conditioning and System Closed-Loop (Signal Conditioning)

- **Industrial Pain Point:** The bridge output U_o is extremely weak, typically only at the millivolt (mV) level. It is highly susceptible to noise submergence and cannot be identified by subsequent ADC stages.
- **Solution:** Introduce an **Instrumentation Amplifier** with a high Common-Mode Rejection Ratio (CMRR) for cascade amplification.
- **Final System Output Formula:**

$$V_{final} = Gain(K) \times U_o$$

Exercise

Description:

A cylindrical load cell made of steel ($E = 2 \times 10^{11}$ Pa, $\nu = 0.3$) has a cross-sectional area of $A = 2 \times 10^{-3}$ m². It is subjected to an axial compressive load of $F = 200$ kN (2×10^5 N).

Four identical strain gauges (Gauge factor $G = 2.0$, $R = 120\Omega$) are bonded to the cylinder to form a temperature-compensated differential full-bridge:

- Two gauges are placed **longitudinally** (parallel to the axis).
- Two gauges are placed **transversely** (perpendicular to the axis).

The bridge excitation voltage is $U_i = 10V$.

Questions:

1. Calculate the longitudinal strain (ϵ_L) and transverse strain (ϵ_T).
2. Calculate the exact output voltage (U_o) of this full-bridge circuit.
3. If the data acquisition system requires a final output voltage of $V_{final} = 5V$, what must be the gain (K) of the instrumentation amplifier?

Solution

Part 1: Mechanical Strain Calculation

- Axial Stress (σ):

$$\sigma = \frac{-F}{A} = \frac{-2 \times 10^5 \text{ N}}{2 \times 10^{-3} \text{ m}^2} = -1 \times 10^8 \text{ Pa}$$

- Longitudinal Strain (ϵ_L) [Compression]:

$$\epsilon_L = \frac{\sigma}{E} = \frac{-1 \times 10^8}{2 \times 10^{11}} = -500 \times 10^{-6} \text{ } (-500\mu\epsilon)$$

- Transverse Strain (ϵ_T) [Tension via Poisson effect]:

$$\epsilon_T = -\nu \cdot \epsilon_L = -0.3 \times (-500\mu\epsilon) = +150 \times 10^{-6} \text{ } (+150\mu\epsilon)$$

Part 2: Bridge Output Voltage (U_o)

- Relative Resistance Changes:

- Longitudinal gauges: $\frac{\Delta R_L}{R} = G \cdot \epsilon_L = 2.0 \times (-0.0005) = -0.001$

- Transverse gauges: $\frac{\Delta R_T}{R} = G \cdot \epsilon_T = 2.0 \times (+0.00015) = +0.0003$

- Bridge Equation:** (Transverse and Longitudinal gauges are placed in adjacent arms)

$$U_o = \frac{\frac{\Delta R_T}{R} - (\frac{\Delta R_L}{R}) + \frac{\Delta R_T}{R} - (\frac{\Delta R_L}{R})}{4} U_i \quad U_o = \frac{0.0026}{4} \times 10V = \mathbf{0.0065V \text{ (6.5mV)}}$$

Part 3: Amplifier Gain (K)

$$V_{final} = K \cdot U_o \Rightarrow K = \frac{V_{final}}{U_o}$$

$$K = \frac{5V}{0.0065V} \approx \mathbf{769.2}$$